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## NPS/003/021 – Technical Specification for CMR Transformers.

### 1. Purpose

This specification details the technical requirements for Continuous Maximum Rated (CMR) system transformers for use on the CE Electric UK distribution networks.

This document supersedes the following documents, all copies of which shall be destroyed.

| Ref         | Version | Title                                        |
|-------------|---------|----------------------------------------------|
| NPS/003/021 | 1.0     | Technical Specification for CMR Transformers |
|             |         | NEDL Turnkey - Transformer CMR Specification |

### 2. Scope

This specification details the technical requirements for CMR transformers for use on the CE Electric UK 33kV, 66kV and 132kV network.

Also included are the technical specifications for neutral displacement voltage transformers (NVDT) and tank mounted auxiliary transformers.

Earthing / auxiliary transformers for use with 66kV and 132kV CMR transformers are specified in NPS/003/026: Technical specification for earthing / auxiliary transformers for use in 66kV and 132kV substations.

This specification does not cover Continuous Emergency Rated (CER) transformers.

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### 3. Technical Requirements

#### 3.1 Overview

CMR transformers may be installed in the following substation types:

- Indoor substations.
- Outdoor substations where exposed HV conductors are not permitted
- Outdoor situations where exposed HV connections are required

#### 3.2 Technical Specification

##### 3.2.1 Overall

The equipment shall comply fully with the Energy Networks Association Technical Specification ENA TS 35-3: Continuous Maximum Rated (CMR) System Transformers (for use on systems up to 132kV) Issue 1, unless varied by this specification in which case this specification shall take precedence.

The equipment shall also comply fully with the current versions of all other relevant, IEC International Standards, British Standard Specifications or equivalent Euro-Norms, and Energy Networks Association Technical Specifications, except where varied by this standard.

The equipment must comply with all relevant UK health and safety legislation.

##### 3.2.2 Variations to ENATS 35-3

The principles of ENATS 35-3 shall also be applied to CMR transformers with a secondary voltage of 20kV, and to CMR transformers with specified power ratings different to those in ENATS 35-3.

Where appropriate the following variations, additions and clarifications to ENATS 35-3 issue 1 are referenced to the clause numbers in that document:

4.2 Loading Cycle: A graph of CMR rating versus ambient temperature and ONAN rating versus ambient temperature shall be provided to CE Electric UK for each transformer at tender stage. These graphs shall be verified against temperature rise test results when these tests are undertaken as described in 10.7 of this document.

5.4 Specification of Tapping Range in Enquiry and Order: The tapping range shall be as specified in Appendix-1: Schedule of Requirements.

5.5 Specification of Short-circuit Impedance: Values of maximum and minimum impedance and +/- tolerances shall be as listed in Appendix 1: Schedule of Requirements.

5.6 Load Loss and Temperature Rise: Losses shall be capitalised using the loss capitalisation values in Appendix 1: Schedule of Requirements and shall be applied as described in Appendix 2: Addendum to Supplier.

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6. Connection and Phase Displacement Symbols for Three-phase Transformers: The requirement will be specified in Appendix 1 Schedule of Requirements. The normal vector groups required for use on CE Electric UK's networks are:

| Voltage (kV) | YEDL        | NEDL           |
|--------------|-------------|----------------|
| 33/11.5      | YNyn0       | YNyn6 or Dyn11 |
| 33/20        | --          | Dyn11          |
| 66/11.5      | YNyn0       | Dyn11/Dyn1*    |
| 66/20        | --          | YNyn0          |
| 66/33        | YNyn0       | YNd11/YNd1*    |
| 132/11.5     | YNd11/YNd1* | YNd11/YNd1*    |
| 132/33       | YNd11/YNd1* | YNd11/YNd1*    |
| 132/66       | YNd11/YNd1* | YNyn0          |

\*the transformer shall have vector changeover links

7.1 Rating Plates: Information to be given in all cases:

In addition the rating plate shall state:

The ONAN rating at 30°C ambient temperature

The CMR rating at 30°C ambient temperature

The nominal tap position expressed in the format: (+/- .... X .... %) +/- .... % in ..... steps

8.2 Oil preservation system: Where the oil preservation system is a free breathing system the breathing pipe shall be terminated with a 1" (one inch) BSP thread to allow the use of a cartridge system of desiccant filters, such as the Brownell Type W system or equivalent. The pipework shall be arranged to allow these cartridges to be changed easily from ground level. These requirements apply to all oil systems that form part of the transformer installation.

10.1 Tests: General: Mechanical tests shall be completed before any electrical type, or routine, tests.

10.1 Tests: General: Further to Table 2, equipment with a nominal system voltage of 11kV shall have a rated lightning impulse voltage of 95kV and a rated 50Hz withstand voltage of 38kV.

10.1.1 Routine Tests: Further to item (i) the two frequency response analysis (FRA) tests shall be carried out on every CMR transformer with input voltages at 66kV and above. The first test shall be done in the factory after all other routine tests and before the transformer is moved or dismantled. The second test shall be done on-site after off-loading and installation/assembly but before on-site commissioning testing.

A report of the FRA tests (including details of the FRA methodology, test criteria, test results, comparison of test results, conclusions and recommendations) shall be supplied to CE Electric UK.

10.1.2 Type tests: For any unit which is to be connected to the CE Electric UK network, the full range of type tests (a) to (h) shall be performed on at least the first unit of a given type and rating from a given production facility. This requirement may be waived if CE Electric UK has had sufficient experience of transformers from the factory and the factory can supply type test criteria from similar units supplied to other UK Distribution Network Operators, or National Grid.

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10.7 Measurement of zero-sequence impedance(s) on three-phase transformers: If in order to accommodate network requirements the measurement criteria need to be varied this will be specified in the Appendix 1 Schedule of Requirements.

10.9 Temperature rise test: A graph of ONAN rating versus ambient temperature and a graph of CMR rating versus ambient temperature shall be produced for every transformer. Copies of these graphs shall be supplied to CE Electric UK. The graphs shall be verified against temperature rise test results during type and/or routine testing.

10.13 Site tests: Tests (i) to (xi) inclusive, and test (xiii) shall be performed on every unit. Where WTI are not being supplied then test (v) is excluded.

12.2 Cooling medium: Where transformers contain insulating fluid then this fluid shall comply fully with CE Electric UK Technical Specification NPS/003/019 – Technical Specification for Electrical Insulating Fluids for use in Plant & Switchgear. This requirement also applies to any fluid used for the first fill and for flushing of the unit.

12.4 Anti-vibration mountings: Anti-vibration mountings shall be matched and tuned to each individual transformer and cooler. The anti-vibration mountings shall be selected to match the longevity of the item they are supporting taking into account likely environmental conditions such as exposure to oil, sunlight etc. Details of the decisions and calculations supporting the anti-vibration measures shall be supplied to CE Electric UK.

13.1 Tanks and covers: Working at Height Regulations: The requirements in ENATS 35-3 shall apply. Details of the provisions incorporated into the design to comply with the Working at Height Regulations must be agreed with CE Electric UK. Where work at height is unavoidable and access is for short time work (This is likely to include: access to the Buchholz relay, winding temperature indicator thermocouples and connections and CT terminations) then access will be gained by portable ladder. To facilitate this access; restraining brackets shall be incorporated into the transformer steelwork to locate and secure the ladder at the top and to allow securing of the ladder at the bottom.

Ladders used by CE Electric UK conform to BS 1129 and BS EN131 and are manufactured from either wood or glass-fibre. The dimensions of ladders used by CE Electric UK are listed below.

|                                                                                     |                   |
|-------------------------------------------------------------------------------------|-------------------|
| Rung diameter (mm)                                                                  | 22 (min) 80 (max) |
| Internal width of section: The minimum distance between sides will not be less than | 235mm.            |

13.2 Surface Finish: The surface finish colour shall be uniform across the transformer and its auxiliaries and shall be a medium neutral grey.

13.3. Terminals: The facings on the transformer need not be arranged to allow for changing from outdoor bushings to cable boxes or vice-versa without any changes to component parts.

13.3 Terminals: HV neutral: The neutral point of star connected HV windings shall be as specified in Appendix 1: Schedule of Requirements and made available via a flange to BS2562 facing 'C' with its major axis horizontal. It shall be readily available via a HV neutral link to allow testing and be enclosed in a lidded, dry metal box. A drawing of the proposed arrangement should be provided for approval.

13.3.3 Terminals: HV Neutral: When the HV neutral is required to be connected via a Neutral Displacement Voltage Transformer (NDVT), the NDVT shall comply with the technical specification in Appendix 5 of this document.

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13.3. Terminals: LV neutral: Further to the requirements of ENATS 35-3, an accessible isolatable link shall be provided on the neutral point of the LV winding to allow testing. The isolatable link arrangement shall allow the neutral point to be connected either solidly to earth or via a cable to a neutral earthing resistor. The LV neutral link arrangement shall be enclosed in a metal cable box preferably the same one used for the LV cable terminations. A drawing of the proposed arrangement should be provided for approval.

13.3.1 Cable Boxes and Unfilled Cable Enclosures: HV terminals: Where separable connectors are used, cable disconnecting links shall be provided in a segregated oil filled chamber as described in ENATS 35-3 clause 13.3.1.

13.3.1 Cable Boxes and Unfilled Cable Enclosures: LV terminals: Where a fully air insulated cable box is supplied, minimum air clearances must comply with the table below:

Fully Air Insulated Cable Box Minimum Clearances

| Voltage (kV) | Live Metal to Earth Flashover distance (mm) | Air end Creepage Distance (mm) | Phase to Phase Clearance (mm) |
|--------------|---------------------------------------------|--------------------------------|-------------------------------|
| 12           | 200                                         | 406                            | 250                           |
| 24           | 220                                         | 550                            | 250                           |
| 36           | 380                                         | 1116                           | 430                           |

13.3.1 Cable Boxes and Unfilled Cable Enclosures: HV and LV terminations: Where cable connections are provided then a suitable supporting bracket equipped with cable cleats shall be provided for the type and number of cable specified in the Appendix 1: Schedule of Requirements.

13.3.2 Cable Termination Chambers: HV and LV Terminations: Where cable connections are required then separable plug-in cable terminations are preferred. Terminations shall comply with CE Electric UK technical specifications:

NPS/002/015 Technical Specification for 11kV and 20kV Joints and Terminations

NPS/002/016 Technical Specification for 33kV Joints and Terminations

NPS/002/017 Technical Specification for 66kV and 132kV Cable Accessories

13.3.2 Cable Termination Chambers: HV and LV Terminations: Where outer cone separable connectors are used these shall be housed in an earthed metal enclosure that:

- (i) When installed on the switchgear provides physical protection, tested to 5J impact at the most vulnerable point on all exposed sides and at the vulnerable point on each exposed edge.
- (ii) Provides a minimum IP rating of IP21B in accordance with IEC 60529. For the purposes of this test the unscreened insulation is to be classed as a hazardous part that requires 20mm clearance from the end of the IP21B test probe.
- (iii) Is ventilated so that it does not create, or encourage, a micro-climate that might lead to partial discharge activity, deterioration or corrosion of the enclosure and its contents.

13.3.2 Cable Termination Chambers: LV terminations: Cable disconnecting links shall be provided. These shall be contained in a separate oil filled chamber such that the links can be removed without draining any oil from the main oil system.

13.3.2 Cable Termination Chambers: LV terminals: Where outer cone cable connectors are used they shall not be stacked more than two per bushing. There shall be the same number of connectors on each bushing.

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13.3.2 Cable Termination Chambers: Gland plates shall be split across each row of gland holes to ease assembly.

13.3.3 Outdoor bushings: All bushings shall be fitted with co-ordinating gaps set as specified in Table 9 of ENATS 35 – 3.

13.3.4 Earthing / Auxiliary Transformer: When required for use with 66kV and 132kV transformers earthing transformers shall comply with CE Electric UK technical specification NPS/003/026.

13.3.4 Unit auxiliary transformer: The positive sequence impedance voltage between the higher voltage winding and the lower voltage (auxiliary) winding shall not be more than 3.0% on 200kVA rating.

13.3.4 Unit auxiliary transformer: Where a tank mounted auxiliary transformer is specified it shall comply with Appendix 6 of this specification and all relevant parts of IEC 60076. Where the main transformer has a secondary voltage of 20kV the mounting flange for the auxiliary transformer shall have a facing of type J (Fig 24 of BS2562).

13.3.4 Unit auxiliary transformer: Where a tank mounted auxiliary transformer is specified the option of cable connected unit will be considered by CE Electric UK. Any agreed option will need to satisfy CE Electric UK protection policy requirements and accessibility requirements.

13.4.1 On Load Tapchanger: Operating mechanism: The term 'Raise' shall mean raise tap number and raise LV volts; the term 'Lower' shall mean lower tap number and lower LV volts.

13.4.2 On Load Tapchanger: Segregation of compartments: The oil-actuated relay provided for the tapchanger shall also operate in the event of loss of oil from the system

13.4.2 On Load Tapchanger: Segregation of compartments: All breather pipework shall comply with the connection requirements of clause 8.2 of this specification.

13.6.2 Cooling Plant: Contactors controlling the power supply to the fans and/or pumps, which are operated from the winding temperature indicator panel in the local substation control room, shall be 110V AC. This supply shall be provided from a 240/55-0-55V transformer in the tap changer.

13.6.2 Cooling Plant: If the transformer cooling system does not have an oil pump and if there are four or more cooling fans then, in order to give increased flexibility in the operation of the cooling system, the operation of half of the cooling fans shall be controlled by what would have been the oil pump control circuit.

13.6.2 Cooling plant: Temperature related variable speed fan motors are preferred to avoid any step increases in noise-level.

13.6.3 Gas-and-oil actuated relay: All valves and relay test facilities shall be padlockable using CE Electric UK standard padlocks.

13.6.3 Gas and Oil Actuated Relay: In addition the gas and oil-actuated relay shall also operate in the event of loss of oil from the system.

13.6.5 Winding Temperature Indicators: Appendix 1 will specify whether a capillary winding temperature indicator or a thermocouple winding temperature sensor is required.

13.6.5 Winding temperature indicators: The type and number of winding temperature indicators required and the windings to which they are to be fitted shall be as specified in Appendix 1: Schedule of Requirements.

13.6.5 Winding temperature indicators; capillary type:

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When capillary type winding temperature indicators are specified the remote indicators shall each have four sets of independently adjustable contacts to give separate control of fan(s) and pump(s) in addition to alarm and trip facilities.

The remote winding temperature indicators shall each provide an analogue facility comprising of a 0 to 10mA DC isolated analogue output for input into the CE Electric UK SCADA system. This output shall be capable of supplying a 1500 ohms load. 0mA shall correspond to 30°C and 10mA shall correspond to 150°C. This facility shall be derived from a linear variable differential transformer mechanically coupled to the winding temperature indicator movement.

#### 13.6.5 Winding temperature indicators; thermocouple sensor type:

When thermocouple winding temperature sensors are specified, temperature measuring thermometers (PT100) shall be located in the top of the transformer. The output from the thermometers shall be terminated in weatherproof compartment using connectors capable of accepting a screened multi-core cable from the winding temperature indicator panel located in the substation control room.

13.6.5 Winding temperature indicators: thermocouple sensor type: Thermocouples shall be wired as shown in CE Electric UK drawing Y030S6701.

13.6.6 Current Transformers: Winding Temperature Indicators: Where a capillary type winding temperature indicator is specified a current transformer shall be provided on the centre termination of the appropriate winding to supply the thermometer pocket heater.

13.6.6 Current Transformers: Winding Temperature Indicators: Where a thermocouple sensor type is specified a current transformer shall be installed on the LV and / or HV winding as specified in Appendix 1, and should be connected as shown in drawing Y030S6701. The current transformer on the LV winding shall supply both the winding temperature indicator and the automatic voltage control schemes. The current transformers shall comply with the following specification.

(i) HV winding temperature monitoring system: A current transformer (5VA Class 3 and ratio 492/1 at 90MVA or 656/1 at 120MVA) shall be provided on the centre phase of the HV winding. This CT shall be terminated in the transformer marshalling cubicle.

(ii) The LV winding temperature monitoring system and voltage control: A current transformer (15VA Class 3 and ratio 1575/1 at 90MVA or 2100/1 at 120MVA) also providing input into the voltage control scheme shall be provided on centre phase of the LV winding. This CT shall be terminated in the transformer marshalling cubicle.

13.6.6 Current Transformers: Automatic voltage control: Where capillary type winding temperature indicators are specified a separate current transformer shall be provided on one terminal of the transformer LV winding to supply the substation automatic voltage control scheme. The secondary connections of this current transformer should be terminated in the transformer marshalling kiosk.

Where a thermocouple sensor type winding temperature indicator is used a common current transformer shall supply it and the automatic voltage control as shown in drawing Y030S6701

13.6.6 Current transformers: Protection scheme: Where site requirements dictate then the transformer shall incorporate CTs for use in protection schemes. These CTs shall be specified in Appendix 1: Schedule of Requirements.

13.6.6 Current Transformers: Test loops: Test loops for current transformers shall be provided generally in accordance with Figure 2 of ENATS 35-3

13.6.6 Current Transformers: Wedding ring current transformers for monitoring fans and pumps: Where thermocouple sensor type winding temperature indicators are specified a minimum of two wedding ring CTs (15/0.015A and 0.05VA) shall be provided for fan and pump current monitoring. These shall be connected in accordance with CE Electric UK drawing Y030S6701.

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13.6.7 Other fittings: All breather pipework shall comply with the connection requirements of clause 8.2 above.

14.1 Drawings: Drawing item (ii) shall detail the oil flow, valve locations and valve functions.

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## 4. References

### 4.1 External Documentation

| Reference             | Title                                                                                |
|-----------------------|--------------------------------------------------------------------------------------|
| ENATS 35-3<br>Issue 1 | Continuous Maximum Rated (CMR) System Transformers (for use on systems up to 132kV), |
| IEC 60076 series      | Power Transformers                                                                   |

### 4.2 Internal documentation

| Reference   | Title                                                                                                                    |
|-------------|--------------------------------------------------------------------------------------------------------------------------|
| NPS/002/015 | Technical Specification for 11kV and 20kV Joints and Terminations                                                        |
| NPS/002/016 | Technical Specification for 33kV Joints and Terminations                                                                 |
| NPS/002/017 | Technical Specification for 66kV and 132kV Cable Accessories                                                             |
| NPS/003/019 | Specification for Electrical Insulating Fluids for use in CE Electric UK Plant & Equipment.                              |
| NPS/003/026 | Technical Specification for Earth/Auxiliary Transformers for use at 132kV and 66kV Substations.                          |
| Y030S6701   | Supply point TX FDR/TX Transformer control Ashridge 852 Plus-P (panel mounted) HV and LV winding temperature indicators. |

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## 4.3 Amendments from Previous Version

| Section | Subject                          | Amendments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2.      | Scope                            | References to NEDL and YEDL replaced by CE Electric UK                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 2.1     | Contents                         | Section added                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 3.2     | Technical Specification          | All references are now made to the issued version of ENATS 35-3 issue 1 rather than the draft version<br>4.2 Graphs of CMR and ONAN versus ambient required at tender stage<br>5.5 Confirmation of method of capitalisation of load losses<br>6. Neutrals added to vector groups<br>6. Requirements for changeover links on 33kv TX removed<br>7.1 Rating Plate: Additional information required stated<br>8.2 Exclusion of refrigerator type breather removed<br>10.1 Requirement for 1/3 octave sound frequency test removed<br>10.1.2 Type tests. Conditions stated for waiving type-test requirement<br>13.1 Requirements with regard to Working at Height Regulations stated<br>13.2 Surface Finish: Colour added.<br>13.3 Requirement to bring neutral point out to disconnecting link added<br>13.3.3 Option for HV NDVT added. Spec added to as Appendix 5<br>13.3.1 Clearances for unfilled cable box added<br>13.3.2 References to NPS specifications for cable terminations added<br>13.3.2 Requirements added with regard to outer cone cable connectors<br>13.3.2 Requirements for cable gland plates added<br>13.3.3 Outdoor bushing: Requirements for co-ordinating gaps added<br>13.3.4 Earthing transformer: Reference made to NPS specification<br>13.3.4 Tank mounted auxiliary transformer: Requirement clarified.<br>13.3.4 Requirement of weir around in-tank tap changer removed<br>13.6.2 Cooling plant: Option of variable speed motors added<br>13.6.3 Gas-and-oil relay: Requirement for loss of oil protection<br>13.6.5 WTI: Requirements for Capillary / thermocouple types clarified<br>13.6.6 CTs: Requirements for WTI and AVC stated<br>13.6.6 CTs: Requirements for test loops stated |
| 3.3     | Amendments from Previous Version | Section added                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|         | Appendix 3                       | Self certification form taken from issued version of ENATS 35-3 Issue 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|         | Appendix 6                       | Specification for tank mounted auxiliary transformer: Added                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|         | Appendix 7                       | Specification for neutral displacement voltage transformer: Added                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

## 5. Definitions

| Term | Definition |
|------|------------|
| NONE |            |

|                           |   |                        |        |             |    |           |    |
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## 6. Authority for issue

### 6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

|           |                    |           |            |
|-----------|--------------------|-----------|------------|
|           |                    | Sign      | Date       |
| K Midgley | Standards Engineer | K Midgley | 23/11/2009 |

### 6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

|             |                         |             |            |
|-------------|-------------------------|-------------|------------|
|             |                         | Sign        | Date       |
| M Nicholson | Head of System Strategy | M Nicholson | 10/12/2009 |

### 6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

|              |                                        |              |            |
|--------------|----------------------------------------|--------------|------------|
|              |                                        | Sign         | Date       |
| D Blackledge | Plant and Switchgear Standards Manager | D Blackledge | 23/11/2009 |

### 6.4 DBD Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

|           |                   |           |            |
|-----------|-------------------|-----------|------------|
|           |                   | Sign      | Date       |
| S Johnson | DBD Administrator | S Johnson | 15/12/2009 |

### 6.5 Approval

Approval is given for the content of this document

|              |                                  |              |            |
|--------------|----------------------------------|--------------|------------|
|              |                                  | Sign         | Date       |
| C Holdsworth | Standards and Assessment Manager | C Holdsworth | 11/12/2009 |

### 6.6 Authorisation

Authorisation is granted for publication of this document

|          |                     |          |            |
|----------|---------------------|----------|------------|
|          |                     | Sign     | Date       |
| C Morgan | Head of Procurement | C Morgan | 11/12/2009 |

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## Appendix 1 - Schedule of Requirements

| Clause<br>ENATS35-3 | Description                                                                                                                                                                                                                                                                                                        | Requirements                                                               |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| 4.1                 | <b>Rated Power</b> MVA<br><i>Tertiary Rated Power (if applicable)</i> MVA<br><b>No-load Voltage Ratio</b> kV<br><i>Alternative voltage ratio</i>                                                                                                                                                                   |                                                                            |
| 5.4                 | Tapping Range                                                                                                                                                                                                                                                                                                      |                                                                            |
| 5.5                 | Short Circuit Impedance % at CMR<br><br>On Principal tap<br><br>On Minimum Tap<br>Maximum Impedance %<br>Minimum Impedance %<br><br>On Maximum Tap<br>Maximum Impedance %<br>Minimum Impedance %<br><br>Short circuit impedance HV/Tertiary<br>At rating MVA<br><br>For Dual LV windings<br>LV1 to LV2 Impedance % |                                                                            |
| 5.6                 | Loss Capitalization Values<br><br>(i) No-load Loss £/kW<br><br>(ii) Load Loss at 50% CER rating £/kW                                                                                                                                                                                                               |                                                                            |
| 6                   | Vector group                                                                                                                                                                                                                                                                                                       |                                                                            |
| 6                   | Vector Links required<br><br>If yes; state alternative vector group required                                                                                                                                                                                                                                       | * Yes / No                                                                 |
| 8.1                 | Neutral conductor and terminal intended to carry load                                                                                                                                                                                                                                                              | * Yes / No                                                                 |
| 8.2                 | Oil preservation system<br><br>If free breathing, breather type                                                                                                                                                                                                                                                    | * Free breathing /<br>Diaphragm conservator<br>* Desiccant / refrigerating |
| 8.3                 | Generator transformer load rejection                                                                                                                                                                                                                                                                               | * Yes / No                                                                 |
| 10.1.1              | FRA test required at works and on site                                                                                                                                                                                                                                                                             | * Yes / No                                                                 |

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| Clause<br>ENATS35-3 | Description                                                                                                                                                                                        | Requirements                           |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| 10.1.3              | Short circuit test required<br>Third harmonic test required                                                                                                                                        | * Yes / No<br>* Yes / No               |
| 10.9                | Temperature rise test required for overload condition                                                                                                                                              | * Yes / No                             |
| 10.10               | Sound power level ONAN dB(A)<br>Sound power level ONAN dB(A)                                                                                                                                       |                                        |
| 12.6                | Maximum HV system short circuit capacity MVA<br>Maximum LV system short circuit capacity MVA                                                                                                       |                                        |
| 13.1                | Transformer to be suitable for sound attenuation enclosure<br>Access through enclosure roof permitted                                                                                              | * Yes / No<br>* Yes/ No                |
| 13.2                | Finish Colour Required                                                                                                                                                                             | Medium neutral grey                    |
| 13.3                | Type of HV Terminations<br>Type of LV Terminations<br>Type of tertiary terminations<br>Type of HV neutral termination<br>Type of LV neutral termination                                            |                                        |
| 13.3.1              | Number and size of single core cables for cable termination<br>For HV Terminations<br>For LV Terminations<br>For tertiary terminations<br>For HV neutral termination<br>For LV neutral termination |                                        |
| 13.3.3              | Bushing stem material                                                                                                                                                                              |                                        |
| 13.3.3              | Co-ordinating gaps required<br>Surge arresters required<br>Brackets for surge arresters required                                                                                                   | * Yes / No<br>* Yes / No<br>* Yes / No |

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| Clause<br>ENATS35-3 | Description                                                                                                                                                                                                                                                                                             | Requirements                                                                     |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| 13.3.4              | Earthing / Auxiliary transformer required<br><br>Free standing or tank mounted<br><br>No-load voltage ratio<br><br>CMR rating kVA<br><br>Zero sequence impedance Ohms/phase<br><br>HV rated short-time neutral current (if not standard) A<br><br>HV terminations<br><br>Vector Group (if not standard) | * Yes / No<br><br><br><br><br><br><br><br><br><br>* Outdoor bushings / cable box |
| 13.4.1              | Max / min tap position indicator required                                                                                                                                                                                                                                                               | * Yes / No                                                                       |
| 13.6.2              | Separate cooler bank required                                                                                                                                                                                                                                                                           | * Yes / No                                                                       |
| 13.6.5              | Winding temperature indicators (WTI)<br><br>Capillary or thermocouple sensors type.<br>WTI on HV winding<br>WTI on LV winding                                                                                                                                                                           | <br><br><br>Yes/No<br>Yes/No                                                     |
| 13.6.6              | Winding temperature indicator CT position HV<br>LV<br>Tertiary<br><br>Additional current transformer accommodation required:<br><br>CTs for voltage control Details:<br>CTs for protection schemes Details:                                                                                             | <br><br><br><br><br>* Yes / No<br><br><br><br><br>                               |

|                           |   |                        |        |             |    |           |    |
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## Appendix 2 - Addendum to Supplier

### Loss Capitalisation Calculation:

Lifetime Cost = Purchase price + ((No load loss kW x No load £/kW) + (Load loss kW x Load loss £/kW)).

[Note that the CE Electric UK loss £/kW figures incorporate utilisation factor and time span]

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### Appendix 3 - Self-Certification Conformance Declaration

CLAUSE BY CLAUSE CONFORMANCE WITH ENA TS 35-3

Transformers covered by ENA TS 35-3 shall comply with the latest issues of the relevant International and British Standards. ENA TS 35-3 is intended to amplify and/or clarify the requirements of those standards.

This check sheet identifies the clauses in ENA TS 35-3 and the clauses of IEC 60076-1. The manufacturer shall declare conformance or otherwise, clause by clause, using the following levels of conformance declaration codes.

|                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Conformance declaration codes</b><br>N/A = Clause is not applicable / appropriate to the product<br>Cs1 = The product conforms fully with the requirements of this clause<br>Cs2 = The product conforms partially with the requirements of this clause<br>Cs3 = The product does not conform to the requirements of this clause | <b>Instructions for completion</b> <ul style="list-style-type: none"><li>When Cs2 or Cs3 is entered, the reason for non-conformance shall be entered</li><li>Where options are specified, the remark should identify the option offered by the manufacturer</li><li>The manufacturer should complete the schedule items in the Remarks column. Additional sheets may be referred to if more detail is required.</li></ul> |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

| Appendix 3 - Self-Certification Conformance Declaration |                                          |                              |                              |               |         |
|---------------------------------------------------------|------------------------------------------|------------------------------|------------------------------|---------------|---------|
| Clause / Sub-clause                                     | Requirement                              | Conformance Code IEC 60076-1 | Conformance Code ENA TS 35-3 | Schedule item | Remarks |
| 1                                                       | Scope and service conditions             |                              |                              | N/A           |         |
| 1.1                                                     | Scope                                    |                              |                              |               |         |
| 1.2                                                     | Normal Service Conditions                |                              | N/A                          | N/A           |         |
| 1.2.2                                                   | Provision for Unusual Service Conditions |                              | N/A                          | N/A           |         |
| 2                                                       | Normative references                     |                              |                              | N/A           |         |

|                           |             |                        |        |             |    |              |
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| <b>Clause / Sub-clause</b> | <b>Requirement</b>                          | <b>Conformance Code<br/>IEC 60076-1</b> | <b>Conformance Code<br/>ENA TS 35-3</b> | <b>Schedule item</b>     | <b>Remarks</b> |
|----------------------------|---------------------------------------------|-----------------------------------------|-----------------------------------------|--------------------------|----------------|
| 3                          | Definitions                                 |                                         | N/A                                     | N/A                      |                |
| 3.1                        | General                                     |                                         |                                         |                          |                |
| 3.2                        | Terminals and Neutral Point                 |                                         | N/A                                     | N/A                      |                |
| 3.3                        | Windings                                    |                                         | N/A                                     | N/A                      |                |
| 3.4                        | Rating                                      |                                         | N/A                                     | N/A                      |                |
| 3.5                        | Tappings                                    |                                         | N/A                                     | N/A                      |                |
| 3.6                        | Losses and No-load Current                  |                                         | N/A                                     | N/A                      |                |
| 3.7                        | Short-circuit Impedance and Voltage Drop    |                                         | N/A                                     | N/A                      |                |
| 3.8                        | Temperature Rise                            |                                         | N/A                                     | N/A                      |                |
| 3.9                        | Insulation                                  |                                         | N/A                                     | N/A                      |                |
| 3.10                       | Connections                                 |                                         | N/A                                     | N/A                      |                |
| 3.11                       | Kinds of Tests                              |                                         | N/A                                     | N/A                      |                |
| 3.12                       | Meteorological Data with Respect to Cooling |                                         | N/A                                     | N/A                      |                |
| 4                          | Rating                                      |                                         |                                         | Rated power MVA          |                |
| 4.1                        | Rating                                      |                                         |                                         | No-load voltage ratio kV |                |
| 4.2                        | Loading cycle                               |                                         |                                         | N/A                      |                |
| 4.3                        | Preferred values of rated power             |                                         | N/A                                     | N/A                      |                |

|                           |             |                        |        |             |    |              |
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| <b>Clause / Sub-clause</b> | <b>Requirement</b>                                                                          | <b>Conformance Code<br/>IEC 60076-1</b> | <b>Conformance Code<br/>ENA TS 35-3</b> | <b>Schedule item</b>                                                       | <b>Remarks</b> |
|----------------------------|---------------------------------------------------------------------------------------------|-----------------------------------------|-----------------------------------------|----------------------------------------------------------------------------|----------------|
| 4.4                        | Operation at higher than rated voltage and/or at disturbed frequency                        |                                         |                                         | N/A                                                                        |                |
| 5                          | Tapped windings                                                                             |                                         | N/A                                     | N/A                                                                        |                |
| 5.1                        | Notation of tapping range                                                                   |                                         |                                         |                                                                            |                |
| 5.2                        | Tapping voltage, current, etc                                                               |                                         | N/A                                     | N/A                                                                        |                |
| 5.3                        | Tapping power                                                                               |                                         | N/A                                     | N/A                                                                        |                |
| 5.4                        | Specification of tappings                                                                   |                                         |                                         | Tapping range offered                                                      |                |
| 5.5                        | Specification of short-circuit impedance<br>Short-circuit impedance between HV and Tertiary |                                         |                                         | Graph of impedance vs. tap position<br>Maximum impedance (%) at rating MVA |                |
| 5.6                        | Load loss and temperature rise                                                              |                                         |                                         | Guaranteed load loss at 50% of CER rating kW                               |                |
| 6                          | Connection and phase displacement symbol                                                    |                                         |                                         | Vector group<br>Alternative vector group where links are provided          |                |
| 7                          | Rating plates                                                                               |                                         | N/A                                     | N/A                                                                        |                |
| 7.1                        | Information to be given in all cases                                                        |                                         |                                         |                                                                            |                |

|                           |   |                        |        |             |    |              |
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| Clause / Sub-clause | Requirement                                                      | Conformance Code<br>IEC 60076-1 | Conformance Code<br>ENA TS 35-3 | Schedule item                                                         | Remarks |
|---------------------|------------------------------------------------------------------|---------------------------------|---------------------------------|-----------------------------------------------------------------------|---------|
| 7.2                 | Additional information to be given                               |                                 | N/A                             | N/A                                                                   |         |
| 8<br>8.1            | Miscellaneous requirements<br>Dimensioning of neutral connection |                                 |                                 | Is neutral connection dimensioned for load current?                   |         |
| 8.2                 | Oil preservation system                                          |                                 |                                 | Description of oil preservation system                                |         |
| 8.3                 | Load rejection on generator transformers                         |                                 |                                 | Is the transformer suitable for generator transformer load rejection? |         |
| 9                   | Tolerances                                                       |                                 | N/A                             | N/A                                                                   |         |
| 10<br>10.1          | Tests<br>General requirements                                    |                                 |                                 | N/A                                                                   |         |
| 10.1.1              | Routine tests                                                    |                                 |                                 | N/A                                                                   |         |
| 10.1.2              | Type tests                                                       |                                 |                                 | Are type test results from an identical unit available?               |         |

|                           |             |                        |        |             |    |              |
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**Appendix 3 - Self-Certification Conformance Declaration**

| <b>Clause / Sub-clause</b> | <b>Requirement</b>                                   | <b>Conformance Code<br/>IEC 60076-1</b> | <b>Conformance Code<br/>ENA TS 35-3</b> | <b>Schedule item</b>                                                                                        | <b>Remarks</b> |
|----------------------------|------------------------------------------------------|-----------------------------------------|-----------------------------------------|-------------------------------------------------------------------------------------------------------------|----------------|
| 10.1.3                     | Special tests                                        |                                         |                                         | Guaranteed and expected change of impedance after short-circuit test<br><br>Special tests included in offer |                |
| 10.2                       | Measurement of winding resistance                    |                                         | N/A                                     | N/A                                                                                                         |                |
| 10.3                       | Measurement of winding ratio and phase displacement  |                                         | N/A                                     | N/A                                                                                                         |                |
| 10.4                       | Measurement of short-circuit impedance and load-loss |                                         |                                         | N/A                                                                                                         |                |
| 10.5                       | Measurement of no-load loss and current              |                                         | N/A                                     | Guaranteed no-load loss kW                                                                                  |                |
| 10.6                       | Measurement of harmonics                             |                                         | N/A                                     | N/A                                                                                                         |                |
| 10.7                       | Measurement of ZPSI                                  |                                         |                                         | Expected ZPSI at 10% flc Ω/ph<br><br>Expected ZPSI at flc Ω/ph                                              |                |
| 10.8                       | Tests on OLTC                                        |                                         | N/A                                     | N/A                                                                                                         |                |
| 10.9                       | Temperature rise test                                | N/A                                     |                                         | N/A                                                                                                         |                |

|                           |             |                        |        |             |    |              |
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| Clause / Sub-clause | Requirement                        | Conformance Code<br>IEC 60076-1 | Conformance Code<br>ENA TS 35-3 | Schedule item                                                                                   | Remarks |
|---------------------|------------------------------------|---------------------------------|---------------------------------|-------------------------------------------------------------------------------------------------|---------|
| 10.10               | Determination of sound levels      | N/A                             |                                 | Guaranteed sound power level (ONAN)<br>dB(A)<br><br>Guaranteed sound power level (CMR)<br>dB(A) |         |
| 10.11               | Pressure test                      | N/A                             |                                 | Maximum permanent deflection<br>mm                                                              |         |
| 10.12               | Frequency Response Analysis        | N/A                             |                                 | Is frequency response analysis is included in offer?                                            |         |
| 10.13               | Site tests                         | N/A                             |                                 | N/A                                                                                             |         |
| 10.14               | Insulation resistance measurements | N/A                             |                                 | N/A                                                                                             |         |
| 10.15               | Magnetic circuit withstand test    | N/A                             |                                 | N/A                                                                                             |         |
| 10.16               | Vacuum test                        | N/A                             |                                 | Maximum permanent deflection<br>mm                                                              |         |
| 11                  | EMC                                |                                 | N/A                             | N/A                                                                                             |         |
| 12                  | Transformer details                | N/A                             |                                 | N/A                                                                                             |         |
| 12.1                | No. of phases and frequency        |                                 |                                 |                                                                                                 |         |

|                           |   |                        |        |             |    |              |
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| <b>Clause / Sub-clause</b> | <b>Requirement</b>          | <b>Conformance Code<br/>IEC 60076-1</b> | <b>Conformance Code<br/>ENA TS 35-3</b> | <b>Schedule item</b>                                       | <b>Remarks</b> |
|----------------------------|-----------------------------|-----------------------------------------|-----------------------------------------|------------------------------------------------------------|----------------|
| 12.2                       | Cooling medium              | N/A                                     |                                         | Cooling medium offered                                     |                |
| 12.3                       | Cooling                     | N/A                                     |                                         | Cooling type at CMR                                        |                |
| 12.4                       | Anti-vibration mountings    | N/A                                     |                                         | Percentage isolation value and description of AV mountings |                |
| 12.5                       | Auxiliary supply voltage    | N/A                                     |                                         | Supply voltage for tap-changer and cooler                  |                |
| 12.6                       | Duty under fault conditions | N/A                                     |                                         | N/A                                                        |                |
| 13                         | Construction details        | N/A                                     |                                         | N/A                                                        |                |

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| <b>Clause / Sub-clause</b> | <b>Requirement</b> | <b>Conformance Code<br/>IEC 60076-1</b> | <b>Conformance Code<br/>ENA TS 35-3</b> | <b>Schedule item</b>                                                                                                                                                                                                                                                                                                                                                                                               | <b>Remarks</b> |
|----------------------------|--------------------|-----------------------------------------|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| 13.1                       | Tanks and covers   | N/A                                     |                                         | Is offer for bolted or welded cover?<br><br>Total oil required (including cooling system) litres<br><br>Filling medium for transport<br><br>Overall height when arranged for transport on trailer mm<br><br>Overall weight when arranged for transport on trailer tonnes<br><br>Total weight as installed in service including cooler plant, all fittings and oil tonnes<br><br>Type of haulage facilities offered |                |
| 13.1.1                     | Covers             | N/A                                     |                                         | N/A                                                                                                                                                                                                                                                                                                                                                                                                                |                |
| 13.1.2                     | Gaskets            | N/A                                     |                                         | N/A                                                                                                                                                                                                                                                                                                                                                                                                                |                |

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| <b>Clause / Sub-clause</b> | <b>Requirement</b>                        | <b>Conformance Code<br/>IEC 60076-1</b> | <b>Conformance Code<br/>ENA TS 35-3</b> | <b>Schedule item</b>                                                          | <b>Remarks</b> |
|----------------------------|-------------------------------------------|-----------------------------------------|-----------------------------------------|-------------------------------------------------------------------------------|----------------|
| 13.2                       | Surface finish                            | N/A                                     |                                         | Finish colour offered                                                         |                |
| 13.3                       | Terminals                                 | N/A                                     |                                         | Type of terminals offered<br>HV<br>LV<br>Tertiary<br>HV neutral<br>LV neutral |                |
| 13.3.1                     | Cable boxes and unfilled cable enclosures | N/A                                     |                                         | Are separate disconnecting chambers provided?                                 |                |
| 13.3.2                     | Cable sealing end chambers                | N/A                                     |                                         | N/A                                                                           |                |
| 13.3.3                     | Outdoor bushings                          | N/A                                     |                                         | Stem diameters and lengths<br>Description of surge protection provision       |                |
| 13.3.4                     | Unit auxiliary transformer                | N/A                                     |                                         | N/A                                                                           |                |
| 13.4                       | On-load tap-changer                       | N/A                                     |                                         | N/A                                                                           |                |

|                           |   |                        |        |             |    |              |
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|----------------------------|---------------------------------|-----------------------------------------|-----------------------------------------|----------------------------------------------------------------------------------|----------------|
| 13.4.1                     | Operating mechanism             | N/A                                     |                                         | Description of motor protection<br>Is a max/min tap position indicator included? |                |
| 13.4.2                     | Segregation of compartments     | N/A                                     |                                         | N/A                                                                              |                |
| 13.4.3                     | Method of operation             | N/A                                     |                                         | N/A                                                                              |                |
| 13.5                       | Clearance to exposed conductors | N/A                                     |                                         | N/A                                                                              |                |
| 13.6                       | Fittings                        | N/A                                     |                                         | N/A                                                                              |                |
| 13.6.1                     | Conservator                     | N/A                                     |                                         | Is low oil level alarm included?                                                 |                |

|                           |   |                        |        |             |    |              |
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| <b>Clause / Sub-clause</b> | <b>Requirement</b>             | <b>Conformance Code<br/>IEC 60076-1</b> | <b>Conformance Code<br/>ENA TS 35-3</b> | <b>Schedule item</b>                                                                                                                                                                                                                                                                                                      | <b>Remarks</b> |
|----------------------------|--------------------------------|-----------------------------------------|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| 13.6.2                     | Cooling plant                  | N/A                                     |                                         | Cooling loss            kW<br>Number of pumps<br>Continuous rating of pump motor shaft   kW<br>Starting current of pump motor            A<br>Number of fans<br>Continuous rating of fan motor shaft       kW<br>Starting current of fan motor            A<br>Fan diameter            mm<br>Speed of fans            rpm |                |
| 13.6.3                     | Gas-and-oil actuated relay     | N/A                                     |                                         | N/A                                                                                                                                                                                                                                                                                                                       |                |
| 13.6.4                     | Pressure relief device         | N/A                                     |                                         | N/A                                                                                                                                                                                                                                                                                                                       |                |
| 13.6.5                     | Winding temperature indicators | N/A                                     |                                         | Description of provision of winding temperature indication                                                                                                                                                                                                                                                                |                |

|                           |             |                        |        |             |    |              |
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**Appendix 3 - Self-Certification Conformance Declaration**

| <b>Clause / Sub-clause</b> | <b>Requirement</b>          | <b>Conformance Code<br/>IEC 60076-1</b> | <b>Conformance Code<br/>ENA TS 35-3</b> | <b>Schedule item</b>                                                                                  | <b>Remarks</b> |
|----------------------------|-----------------------------|-----------------------------------------|-----------------------------------------|-------------------------------------------------------------------------------------------------------|----------------|
| 13.6.6                     | Current transformers        | N/A                                     |                                         | Is current transformer accommodation included?<br><br>Are CT's included in purchaser's specification? |                |
| 13.6.7                     | Other fittings              | N/A                                     |                                         | N/A                                                                                                   |                |
| 13.6.8                     | Valves                      | N/A                                     |                                         | N/A                                                                                                   |                |
| 13.7                       | Marshalling/control box     | N/A                                     |                                         | Means of condensation protection<br>Description of security                                           |                |
| 13.8                       | Interconnecting cables      | N/A                                     |                                         | N/A                                                                                                   |                |
| 13.9                       | Magnetic Circuit            | N/A                                     |                                         | N/A                                                                                                   |                |
| 13.10                      | Core and Winding assemblies | N/A                                     |                                         | N/A                                                                                                   |                |
| 13.11                      | Padlocks                    | N/A                                     |                                         | N/A                                                                                                   |                |
| 13.12                      | Condition Monitoring        | N/A                                     |                                         | Is additional condition monitoring specified?                                                         |                |

|                           |   |                        |        |             |    |              |
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**Appendix 3 - Self-Certification Conformance Declaration**

| <b>Clause / Sub-clause</b> | <b>Requirement</b>                               | <b>Conformance Code<br/>IEC 60076-1</b> | <b>Conformance Code<br/>ENA TS 35-3</b> | <b>Schedule item</b>                     | <b>Remarks</b> |
|----------------------------|--------------------------------------------------|-----------------------------------------|-----------------------------------------|------------------------------------------|----------------|
| 14                         | Documentation                                    | N/A                                     |                                         | Proposed maintenance schedule            |                |
| 14.1                       | Drawings                                         | N/A                                     |                                         | Electronic drawing format                |                |
| 14.2                       | Assembly, operation and maintenance instructions |                                         |                                         | Electronic format for instruction manual |                |

|                           |             |                        |        |             |          |
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#### **Appendix 4 - Manufacturers and Places Of Manufacture, Testing and Inspection**

| <b>Manufacturers and Places of Manufacture, Testing and Inspection</b>                                                                                                                                                                                                                                                                                                                                                                                          |                                                              |                     |                             |                                        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|---------------------|-----------------------------|----------------------------------------|
| <b>Item</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Manufacturer's Drawing Number and/or Type Designation</b> | <b>Manufacturer</b> | <b>Place of Manufacture</b> | <b>Place of Testing and Inspection</b> |
| Transformers complete<br>On-load tap-changer equipment<br>Transformer core<br>HV Bushings<br>LV Bushings<br>Neutral bushings<br>Winding conductor<br>Solid insulation kits<br>Tertiary bushings<br>Transformer tank<br>Radiators<br>Pipe work expansion devices<br>Oil<br>Oil valves<br>Oil pumps<br>Oil pump motors<br>Fans<br>Fan motors<br>Dehydrating breather<br>Conservator diaphragm system<br>Gas and oil actuated relay(s)<br>Main tank<br>Tap-changer |                                                              |                     |                             |                                        |

|                           |   |                        |        |             |    |              |
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| <b>Manufacturers and Places of Manufacture, Testing and Inspection</b>                                                              |                                                              |                     |                             |                                        |
|-------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|---------------------|-----------------------------|----------------------------------------|
| <b>Item</b>                                                                                                                         | <b>Manufacturer's Drawing Number and/or Type Designation</b> | <b>Manufacturer</b> | <b>Place of Manufacture</b> | <b>Place of Testing and Inspection</b> |
| Pressure relief device<br>Outdoor marshalling/control box<br>Temperature indicating devices<br>Material for anti-vibration mounting |                                                              |                     |                             |                                        |
| Any deviation from this Schedule shall be notified in writing as soon as possible for the Purchaser's approval.                     |                                                              |                     |                             |                                        |

|                           |             |                        |  |             |          |
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## Appendix 5 - Additional Technical Schedules to be Completed by Manufacturer

| Item | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Details |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
|      | <b>MAGNETIC CIRCUIT</b>                                                                                                                                                                                                                                                                                                                                                                                                                                         |         |
| 1    | Core construction:<br>(a) Taped/Banded/Bolted limbs<br>(b) Taped/Banded/Bolted yokes<br>(c) Number of limbs<br>(d) Number of limbs wound                                                                                                                                                                                                                                                                                                                        |         |
| 2    | Insulation of:<br>(a) Core bolts<br>(b) Core bolt washers<br>(c) Side plates<br>(d) Core laminations                                                                                                                                                                                                                                                                                                                                                            |         |
| 3    | Whether tank or other flux shields are incorporated Yes/No<br>(If "YES" a full description to be supplied with the Tender)                                                                                                                                                                                                                                                                                                                                      |         |
| 4    | Flux density in the core:<br>(a) Maximum value at rated voltage, 50 Hz, normal tap:<br>(i) wound limbs Tesla<br>(ii) yokes Tesla<br>(iii) shields associated with magnetic circuit Tesla<br>(iv) tank shields Tesla<br><br>(b) Maximum value under any condition of voltage and frequency specified in the Schedules in Part D:<br>(i) wound limbs Tesla<br>(ii) yokes Tesla<br>(iii) shields associated with magnetic circuit Tesla<br>(iv) tank shields Tesla |         |
|      | <b>WINDINGS</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                 |         |

|                           |             |                        |  |             |          |
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| Item | Description                                                                                                                                                                                                                                     | Details |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| 5    | Winding types eg interleaved disc, helical etc:<br>(a) HV windings<br>(b) LV windings<br>(c) Tapping windings (as applicable)<br>(d) Winding sequence, ie Core/___/___/___/___                                                                  |         |
| 6    | Arrangement of tappings (Linear, Coarse/Fine, Reversing)                                                                                                                                                                                        |         |
| 7    | Type of tap changer:<br>(a) Neutral end<br>(b) Line end<br>(c) Separate tank design<br>(d) In-tank design in separate tank                                                                                                                      |         |
| 8    | Conductor material for :<br>(a) HV windings<br>(b) LV windings<br>(c) Tapping windings (as applicable)                                                                                                                                          |         |
| 9    | Conductor insulation:<br>(a) HV windings<br>(b) LV windings<br>(c) Tapping windings                                                                                                                                                             |         |
| 10   | Oil circulation (ie Natural/Partially directed/Directed:<br>(a) To the windings:<br>(i) HV windings<br>(ii) LV windings<br>(iii) Tapping windings<br>(b) Through the windings:<br>(i) HV windings<br>(ii) LV windings<br>(iii) Tapping windings |         |
| 11   | Short circuit capability:<br>(a) Potential axial thrust for worst fault condition of each winding:<br>(i) HV windings tonnes<br>(ii) LV windings tonnes                                                                                         |         |



|                           |   |                        |  |             |          |
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| Item | Description                                                                                                                                                                                                                                                                                                                                                                        | Details |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
|      | (b) Impedance on minimum tapping:<br>Tap position number<br>HV/LV %<br>(c) Impedance on maximum tapping:<br>Tap position number<br>HV/LV %                                                                                                                                                                                                                                         |         |
| 19   | Demonstrable zero phase sequence impedance at 75°C and on normal tapping, referred to HV side, at 10% full load current in each phase winding (tertiary terminal open circuit):<br>(a) HV-N (HV- /N) $\Omega$ /phase<br><i>[Connect A-B-C: a-b-c open circuit: supply ABC to YN]</i><br>(b) HV-LV//N $\Omega$ /phase<br><i>[Connect A-B-C: Connect a-b-c-YN: supply ABC to YN]</i> |         |
| 20   | Calculated zero phase sequence impedance at 75°C and on normal tapping, referred to HV side, assuming rated voltage (single phase) applied between line terminals and neutral:<br>(a) $Z_H$ $\Omega$ /phase (approx)<br>(b) $Z_L$ $\Omega$ /phase (approx)                                                                                                                         |         |
| 21   | Hottest spot winding temperature, most onerous tap position (ambient air temperature 30°C):<br>Tap No.<br>(a) At rated power (tertiary windings not loaded):<br>(i) HV °C<br>(ii) LV °C                                                                                                                                                                                            |         |
| 22   | Maximum observable oil temperature (ambient air temperature 30°C):<br>(a) At rated power :<br>(i) top oil °C<br>(ii) at inlet to cooler °C<br>(iii) at outlet from cooler °C<br>(b) At ONAN rating:<br>(i) top oil °C                                                                                                                                                              |         |
| 23   | Anticipated hottest spot winding temperature:<br>(a) Emergency cyclic loading conditions:<br>(i) HV °C<br>(ii) LV °C                                                                                                                                                                                                                                                               |         |

|                           |             |                        |  |             |    |           |    |
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| Item | Description                                                                                                                                                                                                                                                                                                                                                                                                  | Details |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
|      | (b) With 1.5 pu loading for 30 mins. at 20°C ambient temperature with pre-load of 0.7 pu °C                                                                                                                                                                                                                                                                                                                  |         |
| 24   | Percentage of sum of load and no-load losses, at rated power, that will be supplied during temperature rise tests %                                                                                                                                                                                                                                                                                          |         |
| 25   | Calculated winding capacitance:<br>(a) Approximate series capacitance of each phase winding:<br>(i) HV pF<br>(ii) LV pF<br>(b) Approximate shunt capacitance to earth of each phase winding with core and tank earthed:<br>(i) HV to earth with LV winding unearthed pF<br>(ii) LV to earth with LV winding unearthed pF<br>(c) Approximate capacitance HV to LV phase winding with LV winding un-earthed pF |         |
| 26   | Guaranteed sound power level:<br>(a) main unit dB(A)<br>(b) cooler bank dB(A)                                                                                                                                                                                                                                                                                                                                |         |
|      | TANK AND COOLER                                                                                                                                                                                                                                                                                                                                                                                              |         |
| 27   | Tank material                                                                                                                                                                                                                                                                                                                                                                                                |         |
| 28   | Thickness of tank:<br>(a) sides mm<br>(b) base mm<br>(c) cover mm                                                                                                                                                                                                                                                                                                                                            |         |
| 29   | Whether tank base suitable for air cushion equipment                                                                                                                                                                                                                                                                                                                                                         |         |
| 30   | Thickness of radiator plates and/or cooling tubes. mm                                                                                                                                                                                                                                                                                                                                                        |         |
| 31   | Total oil required, including cooling system litres                                                                                                                                                                                                                                                                                                                                                          |         |
| 32   | Volume of oil to be removed to effect in-situ change of:<br>(a) HV bushing litres<br>(b) LV bushing litres<br>(c) Fitting of bushing turret litres                                                                                                                                                                                                                                                           |         |
| 33   | Total volume of conservator(s) litres                                                                                                                                                                                                                                                                                                                                                                        |         |

|                           |             |                        |  |             |    |           |    |
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| Item | Description                                                                                                                                                                                                                                                  | Details |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| 34   | Volume of oil in (each) conservator between highest and lowest visible levels<br>litres                                                                                                                                                                      |         |
| 35   | Height of oil above pad level at maximum level in the conservator(s)<br>metres                                                                                                                                                                               |         |
| 36   | Total number of oil pumps                                                                                                                                                                                                                                    |         |
| 37   | Output of each oil pump under service conditions:<br>(a) quantity<br>litres/sec<br>(b) head<br>metres head of oil (approx)                                                                                                                                   |         |
| 38   | Continuous rating of each oil pump motor shaft<br>kW                                                                                                                                                                                                         |         |
| 39   | Starting current of each pump motor<br>Amps                                                                                                                                                                                                                  |         |
| 40   | Rated normal output of each fan<br>(a) Quantity<br>m <sup>3</sup> /s<br>(b) Head<br>mm water gauge                                                                                                                                                           |         |
| 41   | Total number of fans                                                                                                                                                                                                                                         |         |
| 42   | Nominal diameter of fans<br>mm                                                                                                                                                                                                                               |         |
| 43   | Speed of fans<br>rpm                                                                                                                                                                                                                                         |         |
| 44   | Continuous rating of each fan motor shaft<br>kW                                                                                                                                                                                                              |         |
| 45   | Starting current of each fan motor<br>Amps                                                                                                                                                                                                                   |         |
| 46   | Maximum temperature of tank surface, measured from ground level:<br>(a) at rated power and at 30°C ambient:<br>(i) above 2.6 metres<br>°C<br>Location<br>(ii) below 2.6 metres<br>°C<br>Location<br>(b) under cyclic load at 10°C ambient:<br>°C<br>Location |         |
|      | GENERAL                                                                                                                                                                                                                                                      |         |
| 47   | Filling medium for transport                                                                                                                                                                                                                                 |         |
| 48   | Total weight as installed for service, including cooling plant, all fittings and oil:<br>(a) quotation estimate<br>tonnes                                                                                                                                    |         |

|                           |             |                        |  |             |    |           |    |
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| Item | Description                                                                                                                                                                                                                                                                                                                              | Details |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
|      | (b) final design calculation tonnes                                                                                                                                                                                                                                                                                                      |         |
| 49   | Weight arranged for transport:<br>(a) excluding vehicle tonnes<br>(b) including vehicle tonnes                                                                                                                                                                                                                                           |         |
| 50   | Weight of each individual cooler, including oil tonnes                                                                                                                                                                                                                                                                                   |         |
| 51   | Weight of each bushing insulator:<br>(a) HV kg<br>(b) LV kg<br>(c) Neutral kg                                                                                                                                                                                                                                                            |         |
| 52   | Tanks, conservators and oil filled compartments:<br>(a) permanent deflection after vacuum test:<br>(i) tank mm<br>(ii) conservator mm<br>(iii) other oil filled compartments [specify] mm<br>(b) Permanent deflection after oil pressure test:<br>(i) tank mm<br>(ii) conservator mm<br>(iii) other oil filled compartments [specify] mm |         |

|                           |             |                        |  |             |    |           |    |
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## Appendix 6 – Specification for Transformer Tank Mounted Auxiliary Transformers

### A6.1 Overview

ENATS 35-3 allows for flange mounting of an outdoor auxiliary transformer. The auxiliary transformer shall be designed, constructed and tested in accordance with ENATS 35-1 - Distribution transformers, except where varied by this specification.

### A6.2. Electrical Performance Requirements

|                                          |                                                  |
|------------------------------------------|--------------------------------------------------|
| A6.2.1: Input voltage for equipment (Um) | To match the main transformer secondary voltage. |
| A6.2.2: Output voltage                   | 415V at no load.                                 |
| A6.2.3: Vector Group                     | Dyn11                                            |
| A6.2.4: Rating                           | As specified in the schedule of requirements.    |
| A6.2.5: Tappings                         | Not required                                     |
| A6.2.6: Impedance Voltage (at 75°C)      | 4.5% of rated power                              |

### A6.3. Mounting Arrangements

A6.3.1: Mounting Arrangement: The flange facing and mounting arrangement shall comply with section 13.3.4 of ENATS 35-3 and section 13.3.4 of CE Electric UK technical specification NPS/003/021.

A6.3.2: Mounting Arrangements: The auxiliary transformer shall be rigidly supported by the flange mounting arrangement. Additional bracings shall be provided between the auxiliary and main transformer if necessary to ensure adequate and rigid support.

### A6.4. LV Terminal Arrangements

A6.4.1: An LV fuseboard shall be mounted on the auxiliary transformer and the LV terminals of the auxiliary transformer shall be connected to this LV fuseboard. The means of connection shall minimise any risk from failure of this connection(s).

A6.4.2: The LV fuseboard shall conform fully to Form 4 Type 2 requirements of BS EN 60439-1, Low-voltage switchgear and controlgear assemblies – Type-tested and partially type-tested assemblies. All terminals and exposed live metalwork shall be shrouded.

A6.4.3: The LV fuseboard shall be designed and constructed to be suitable for outdoor use. The fuseboard, its components (excluding insulating parts) and its mountings shall not require maintenance for a period of at least 30 years in a polluted / coastal environment according to EN ISO 12944-2 Category C4.

A6.4.4: The fuseboard housing shall have a padlockable access door/non-bolted cover; suitable for use with a padlock hasp of 3-6mm diameter.

A6.4.5: The auxiliary transformer LV phase connections shall be terminated, in the LV fuseboard, onto three 100A fuse units that shall be in accordance with Type 3 of BS 7657 (1993); each of which shall be fused with a 60A fuse that shall be in accordance with Type II b of BS1361 (1971).

A6.4.7: The auxiliary transformer LV fuses shall be ergonomically accessible from ground level and be mounted not less than 900mm and not more than 1600mm above ground level.

|                           |   |                        |  |             |    |           |    |
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A6.4.8: The auxiliary transformer LV neutral connections shall be terminated onto an arrangement with two independently removable, bolted links inside the LV fuseboard:  
(a) the outgoing side of one link shall be connected to a designated earth connection point on the auxiliary transformer  
(b) the outgoing side of one link shall be connected to the outgoing LV cable

A6.4.9: The underside of the LV fuseboard housing shall have a suitably positioned 20mm diameter cable access hole, equipped with a compression gland for a 4-core armoured 25mm<sup>2</sup> cable that will be used to connect to the outgoing sides of the LV fuses.

A6.4.10: The LV fuseboard shall be equipped with all necessary connections/lugs to allow termination of a 4-core armoured 25mm<sup>2</sup> cable used to connect to the outgoing sides of the LV fuses

## **A6.5. Earthing Terminal Arrangement**

A6.5.1: An earthing connection shall be provided on the exterior of the auxiliary transformer. This connection shall be designed to accept neutral/earth connection(s) from the bolted link in the LV fuseboard described above.

|                           |             |                        |  |             |    |           |    |
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### A6-6. Auxiliary Transformer Conformance Declaration

The conformance declaration table below shall be completed for each unit.

| Clause | Description                                      | Confirmation/Comment/Description |
|--------|--------------------------------------------------|----------------------------------|
| A6.1.1 | Conformance with ENATS 35-1                      |                                  |
| A6.2.1 | Input voltage                                    |                                  |
| A6.2.2 | Output voltage                                   |                                  |
| A6.2.3 | Vector group                                     |                                  |
| A6.2.4 | Rating                                           |                                  |
| A6.2.5 | Tappings                                         |                                  |
| A6.2.6 | Impedance voltage (at 75°C)                      |                                  |
| A6.3.1 | Flange facing                                    |                                  |
| A6.3.2 | Mounting arrangements                            |                                  |
| A6.4.1 | LV terminals & connection                        |                                  |
| A6.4.2 | Form 4 Type 2 requirements of BS EN 60439-1      |                                  |
| A6.4.3 | Withstand 30 years in EN ISO 12944-2 Category C4 |                                  |
| A6.4.4 | Padlockable door/cover                           |                                  |
| A6.4.5 | LV fuse & link arrangements                      |                                  |
| A6.4.6 | Neutral connection                               |                                  |
| A6.4.7 | Cable hole & gland                               |                                  |
| A6.5.1 | Earth terminal arrangement                       |                                  |

|                           |             |                        |  |             |    |           |    |
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## Appendix 7 – Specification for Transformer Tank Mounted Neutral Displacement Voltage Transformer (NDVTs)

### A7-1 Overview

A5.1.1 ENATS 35-2 allows for flange mounting of an outdoor voltage transformer for neutral displacement protection. The NDVT primary winding shall be connected between the transformer winding star point and earth.

A5.1.2 The NDVT shall be designed, constructed and tested in accordance with BS EN 60044:1999 Instrument transformers – Part 2 Inductive voltage transformers

### A7-2. Electrical Performance Requirements

A5.2.1 Highest voltage for equipment (Um) 36 kV

A5.2.2 Secondary winding output voltage 63.5 V

A5.2.3 Minimum rating 50 VA

A5.2.4 Accuracy Class 6P

A5.2.5 Basic Insulation Oil Immersed

### A7-3. General Construction and Fittings

A5.3.1: The oil in the NDVT shall be separated from the oil in the main transformer by means of an internal barrier that shall be pressure tested in accordance with section 10.11 of ENATS 35-2, in both directions.

A5.3.2: The NDVT shall incorporate the following features:

Oil preservation system in accordance with Sub clause 8.2 of section 3.3.2 of NPS/003/012

Tank and covers in accordance with Sub clause 13.1 of section 3.3.2 of NPS/003/012 and within the constraints CE Electric UK drawing number Y213X99312.

Oil level indicator

Rating and diagram plate

Terminal box with removable, padlockable cover. This terminal box shall contain links as required to comply with the winding arrangements specified in section A5-4 below.

These links shall be clearly identified.

A cable gland for a 2 core multicore cable

### A7-4. Winding Arrangements

A5.4.1: The earthing of the primary winding shall be independent of, and segregated from, the earthing of the secondary winding.

A5.4.2: Primary winding terminal "N": The primary winding neutral point shall be provided with an accessible isolatable link to allow testing. The primary winding neutral link should be enclosed in a lidded, dry metal box. A drawing of the proposed arrangement should be provided for approval.

A5.4.3: The primary winding terminal "A" shall be connected to a HV bushing by a flexible lead.

This bushing shall connect the NDVT to the main transformer HV neutral and shall be within the main transformer oil system.

It shall be possible to easily disconnect this flexible lead to the bushing for the purpose of testing the main transformer.

A5.4.4: The secondary winding terminals "a" and "n" shall be connected to links (typically GEC RS20 type) or equivalent in the terminal box. The "n" terminal winding neutral point shall be provided with an accessible isolatable link to allow testing. The secondary winding neutral link should be enclosed in a lidded, dry metal box. A drawing of the proposed

|                           |   |                        |  |  |             |    |           |    |
|---------------------------|---|------------------------|--|--|-------------|----|-----------|----|
| <b>Document reference</b> |   | NPS/003/021            |  |  |             |    |           |    |
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arrangement should be provided for approval.

|                           |             |                        |  |             |    |           |    |
|---------------------------|-------------|------------------------|--|-------------|----|-----------|----|
| <b>Document reference</b> | NPS/003/021 |                        |  |             |    |           |    |
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## A7-5. NDVT Conformance Declaration

The conformance declaration table below shall be completed for each unit.

| Clause | Description                                                                                                             | Confirmation/Comment/Description |
|--------|-------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| A7.1.1 | NDVT primary winding                                                                                                    |                                  |
| A7.1.2 | Compliance with BS EN 60044:1999 Instrument transformers – Part 2 Inductive voltage transformers                        |                                  |
| A7.2.1 | Highest voltage for equipment (Um)                                                                                      |                                  |
| A7.2.2 | Secondary winding output voltage                                                                                        |                                  |
| A7.2.3 | Burden rating                                                                                                           |                                  |
| A7.2.4 | Accuracy class                                                                                                          |                                  |
| A7.2.5 | Basic insulation type                                                                                                   |                                  |
| A7.3.1 | Barrier pressure test                                                                                                   |                                  |
| A7.3.2 | Oil preservation system in accordance with section 8.2 of NPS-003-012                                                   |                                  |
|        | Tank and covers in accordance with section 13 of NPS-003-012 and within the constraints CEEUK drawing number Y213X99312 |                                  |
|        | Oil level indicator                                                                                                     |                                  |
|        | Rating and diagram plate                                                                                                |                                  |
|        | Terminal box with removable, padlockable cover.                                                                         |                                  |
|        | This terminal box links.                                                                                                |                                  |
|        | Cable gland for a 2 core multicore cable                                                                                |                                  |
| A7.4.1 | Independent, segregated primary and secondary earthing                                                                  |                                  |
| A7.4.2 | Primary winding terminal "N"                                                                                            |                                  |
| A7.4.3 | Primary winding terminal "A"                                                                                            |                                  |
| A7.4.4 | Secondary winding terminals "a" and "n"                                                                                 |                                  |